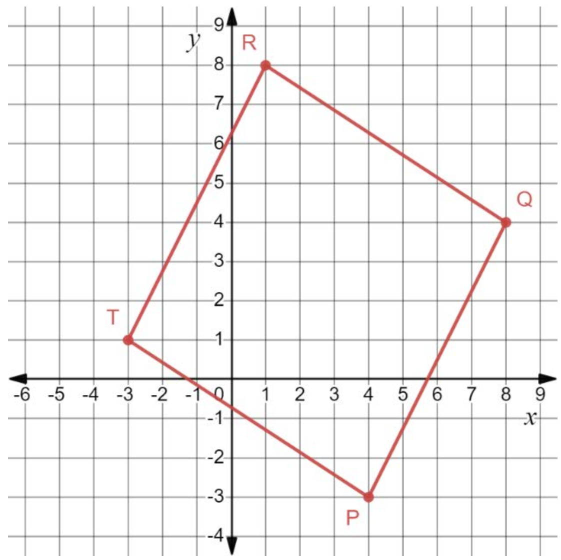
Geometry
Unit 13
Lesson 8 Homework

Name___Answer Key_______
Date____________________
Period__________________

Use the following information for questions 1-4. $PQRT$ has vertices at $P(4,-3)$, $Q(8,4)$, $R(1,8)$, and $T(-3,1)$.

1. Graph $PQRT$ on the coordinate plane.



2. Complete the table.

Line Segment	Length	Slope	Length of Diagonal Line Segment
$\overline{PQ}$	$\sqrt{7^2+4^2}=\sqrt{65}$	$\frac{4-(-3)}{8-4}=\frac{7}{4}$	
$\overline{QR}$	$\sqrt{7^2+4^2}=\sqrt{65}$	$\frac{8-4}{1-8}=-\frac{4}{7}$	
$\overline{RT}$	$\sqrt{7^2+4^2}=\sqrt{65}$	$\frac{8-1}{1-(-3)}=\frac{7}{4}$	
$\overline{PT}$	$\sqrt{7^2+4^2}=\sqrt{65}$	$\frac{1-(-3)}{-3-4}=-\frac{4}{7}$	
$\overline{PR}$			$\sqrt{11^2+3^2}=\sqrt{130}$
$\overline{QT}$			$\sqrt{11^2+3^2}=\sqrt{130}$

3. Circle all classifications that apply to $PQRT$.

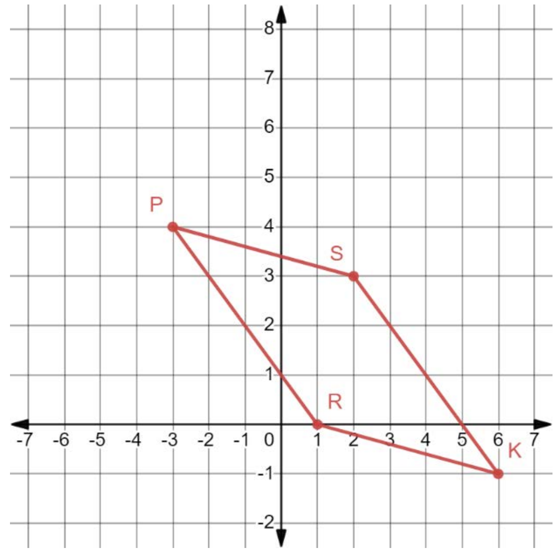
Isosceles Trapezoid   Kite   Parallelogram   Rectangle

Rhombus   Square   Trapezoid

4. Complete the statement.

Since $PQRT$ ✖ has / o does not have congruent diagonals, $PQRT$ is classified as a(n)

square.



Use the following information for question 5. $PRKS$ has vertices at $P(-3,4)$, $R(1,0)$, $K(6,-1)$, and $S(2,3)$.

5. $PRKS$ is classified as a parallelogram because...

$PS = \sqrt{5^2 + 1^2} = \sqrt{26}$

$SK = \sqrt{4^2 + 4^2} = \sqrt{32}$

$RK = \sqrt{5^2 + 1^2} = \sqrt{26}$

$PR = \sqrt{4^2 + 4^2} = \sqrt{32}$

Slopes:

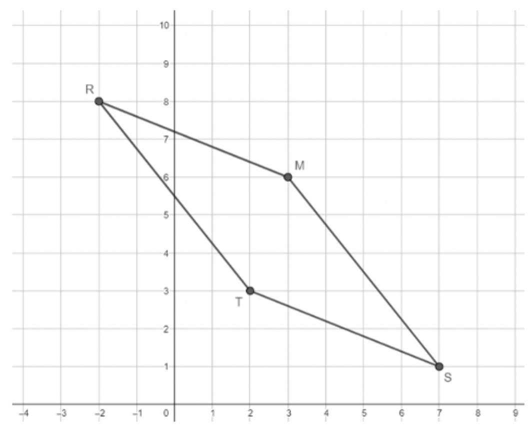
$\overline{PS}: -\frac{1}{5}$

$\overline{SK}: -1$

$\overline{RK}: -\frac{1}{5}$

$\overline{PR}: -1$

Opposite sides, but not all sides, congruent.
Opposite sides parallel.



Use the following information for question 6-7. Saniya graphed $SMRT$ and classified the quadrilateral as a kite.

6. Do you agree with Saniya? No.

$RM = \sqrt{5^2 + 2^2} = \sqrt{29}$

$ST = \sqrt{5^2 + 2^2} = \sqrt{29}$

$SM = \sqrt{5^2 + 4^2} = \sqrt{41}$

$RT = \sqrt{5^2 + 4^2} = \sqrt{41}$

No adjacent sides are congruent, so not a kite.

7. Justify your answer using properties and showing your work.
Side lengths show that opposite sides are congruent, but adjacent sides are not.